

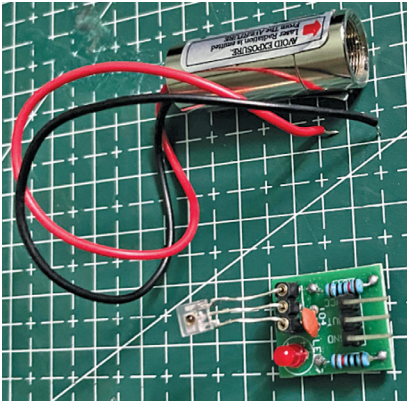


Fig. 4: Laser transmitter (Tx) and receiver module

TABLE 2
BOARD LABELS AND PIN
FUNCTIONS OF THE
MICROCONTROLLERS

Board Pin	Component Pin
We Mos D1 (Rx)	Laser receiver
GND	GND
5V	VCC
I/O Pin 5	OUT
Arduino Uno/We Mos D1 (Tx)	Laser Transmitter
GND	GND
I/O 3	VCC

with ESP8266, nRF with ESP8266, or STM32 with Arduino may also be employed.

Fig. 1 displays the authors' prototype through key visuals, while Fig. 2 outlines the system's block diagram. The components used are listed in the Bill of Materials Table 1.

Circuit and working

The circuit is relatively simple. Fig. 3 shows the combined circuit diagram of transmitter and receiver based on WeMos D1 Mini. On the

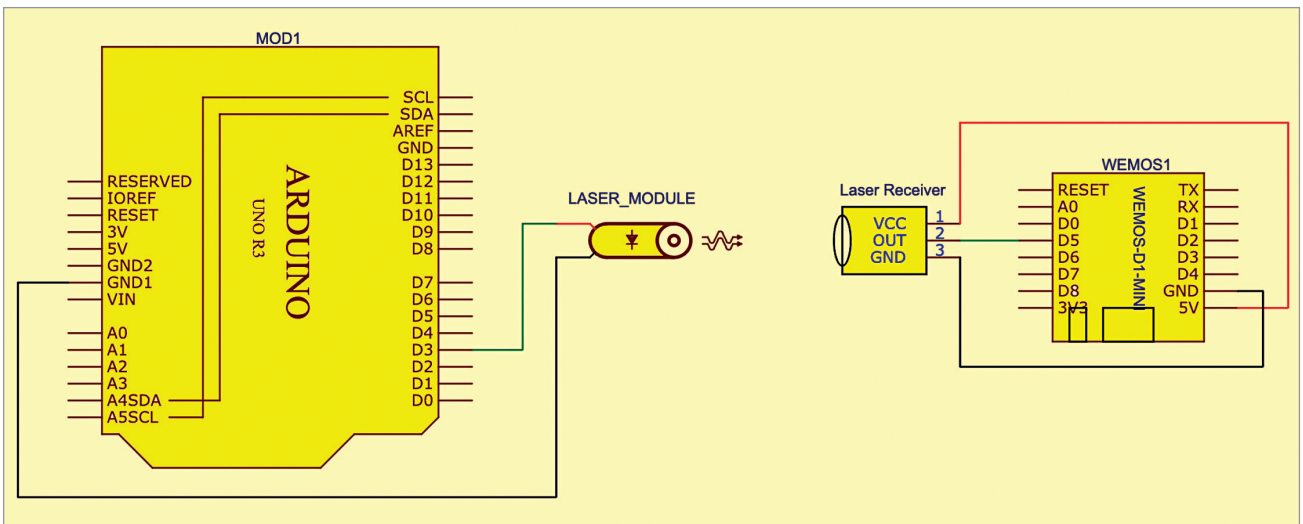


Fig. 5: Alternative circuit

```

1 //Transmitter Code:
2 #define LED_PIN 3
3 #define BITDELAY 3
4 void setup() {
5   pinMode(LED_PIN, OUTPUT);
6   Serial.begin(9600);
7   digitalWrite(LED_PIN, HIGH);
8 }
9 void loop() {
10  String inputString = Serial.readStringUntil('\n'); if (inputString.length() > 0) { Serial.print("Transmitted Text: ");
11  Serial.println(inputString);
12  transmitString(inputString);
13  }
14  }
15  void transmitString(const String &str) {
16  for (int i = 0; i < str.length(); i++) {
17  char currentChar = str.charAt(i);
18  transmit_byte(currentChar);
19  }
20  }
21  void transmit_byte(char byte)
22  {
23  digitalWrite(LED_PIN, LOW); delay(BITDELAY);
24  for (int i = 0; i < 8; i++) {
25  digitalWrite(LED_PIN, (byte & (1 << i)) != 0);
26  delay(BITDELAY);
27  }
28  digitalWrite(LED_PIN, HIGH);
29  delay(BITDELAY);
30  }

```

Fig. 6: Snippet of the transmitter source code

left side is the receiver circuit, and on the right side is the transmitter circuit with the laser diode (see Fig. 3). Refer to Table 2 for board label specifications and pin functions of the microcontrollers.

The objective is to design a customised optical communication link that enables seamless synchronisation between heterogeneous units—the transmitter (Tx) and the receiver (Rx). At the transmitter end, a laser unit is interfaced with an Arduino microcontroller. At the receiver end, a photodiode sensor (ISO 203) is used. This is a compact 3-pin light sensor with an integrated amplifier and an open-collector transistor at its output. Fig. 4 shows the laser